

Higher derivatives



- 1
- a $\frac{dy}{dx} = -10x^{-6}$ b $\frac{d^2y}{dx^2} = 60x^{-7}$ c $\frac{d^3y}{dx^3} = -420x^{-8}$
- 2
- a $\frac{dy}{dx} = 40x^7$ b $\frac{d^2y}{dx^2} = 280x^6$ c $\frac{d^3y}{dx^3} = 1680x^5$ d $\frac{d^4y}{dx^4} = 8400x^4$
- e $\frac{d^5y}{dx^5} = 33\,600x^3$
- 3
- a $\frac{dy}{dx} = -8x^3 + 12x^2 - 14x + 8$ b $\frac{dy}{dx} = -142$
- c $\frac{d^2y}{dx^2} = -24x^2 + 24x - 14$ d $\frac{d^2y}{dx^2} = -302$
- e $\frac{d^3y}{dx^3} = -48x + 24$ f $\frac{d^3y}{dx^3} = -264$
- 4
- a $y'' = 12x^2 - 36x + 16$ b $f''(x) = -\frac{6}{49}x^{-\frac{13}{7}}$
- c $f''(x) = 108(3x + 2)^2$ d $f''(x) = 24(x + 8)^{-4}$
- e $y'' = -4(7 - 6x^2)^{-\frac{5}{3}}(2x^2 + 7)$ f $y'' = -(6 - x)^{-\frac{3}{2}}$
- g $f''(x) = 20x^3 + 6$ h $f''(x) = -84x^5 + 36x^2 - 24x$

Concavity

- 5
- a B, C, D b A
- 6
- a $f'(a) < 0, f''(a) > 0$ b $f'(a) > 0, f''(a) < 0$
- c $f'(a) > 0, f''(a) > 0$ d $f'(a) < 0, f''(a) < 0$

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a

i Positive

ii Minimum

iii $f'(x) = 2x - 4$

iv $f''(x) = 2$

v Concave up

b

i Negative

ii Maximum

iii $f'(x) = -2x + 4$

iv $f''(x) = -2$

v Concave down

8

a $f'(x) = 8x + 3$

b $f''(x) = 8$

c

x	0	1	2	3	4
$f'(x)$	3	11	19	27	35

d Gradient is increasing at a constant rate.

e No, because $f(x)$ is a quadratic function, and it is always concave up.

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a $f'(x) = -6x + 12$

b $f''(x) = -6$

c Gradient is decreasing at a constant rate.

d No, because $f(x)$ is a quadratic function, and it is always concave down.

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a

- i No such x -values
- ii $x > -1$
- iii The graph is always increasing for $x \geq -1$. There is no value of x in the function such that $f''(x) = 0$ and concavity changes.

b

- i $x > 2$
- ii No such x -values
- iii The graph is always decreasing for $x \geq 2$. There is no value of x in the function such that $f''(x) = 0$ and concavity changes.

c

- i $x > 0$
- ii $x < 0$
- iii The graph is always decreasing for $x < 0$ and for $x > 0$. There is no value of x in the function such that $f''(x) = 0$ and concavity changes.

d

- i $x < 0$ or $x > 0$
- ii No such x -values
- iii The graph is always increasing for $x < 0$ and always decreasing for $x > 0$. There is no value of x in the function such that $f''(x) = 0$ and concavity changes.

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a Horizontal translation of 6 units left

b $(0, 0)$ c $(-6, 0)$

d

x	-7	-6	-5
y'	3	0	3
y''	-6	0	6

e Yes, because $f'(-6) = 0$.f $x > -6$

12

a Horizontal translation of 5 units left

b $(0, 0)$ c $(-5, 0)$

d

x	-6	-5	-4
y'	3	0	3
y''	-6	0	6

e Yes, because $f'(-5) = 0$.f $x > -5$

13



a $y'' = 12x^2 - 48x$

b $(0, -9), (4, -265)$

c

x	-2	0	2	4	6
y'	-128	0	-96	-128	0
y''	144	0	-48	0	144

d $(0, -9)$ is a horizontal point of inflection and $(4, -265)$ is an ordinary point of inflection.

e $x < 0$ or $x > 4$

14

a $y'' = 24x - 32$

b $\left(\frac{4}{3}, -\frac{206}{27}\right)$

c An ordinary point of inflection

d $x < \frac{4}{3}$

15

a $y'' = 18x + 12$

b $\left(-\frac{2}{3}, -\frac{5}{9}\right)$

c Ordinary

d $x < -\frac{2}{3}$

16

a $y'' = 20x^3 - 6$

b $\left(\sqrt[3]{\frac{3}{10}}, \left(\sqrt[3]{\frac{3}{10}}\right)^5 - 3\left(\sqrt[3]{\frac{3}{10}}\right)^2\right)$

c Ordinary

d $x > \sqrt[3]{\frac{3}{10}}$

e $x < \sqrt[3]{\frac{3}{10}}$

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a $x < -3$ or $x > 0$

b $-3 < x < 0$

c $f''(x) = 6x + 9$

d $x > -\frac{3}{2}$

e $x < -\frac{3}{2}$

f $x = -3$

g $x = -\frac{3}{2}$

Stationary points

18

a $x = 5$

b $f''(5) = 2$



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- a $(-4, 0), (2, -108)$
- b $f''(x) = 6x + 6$
- c $(-4, 0)$ is a maximum turning point and $(2, -108)$ is a minimum turning point.
- d $x < -1$

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- a $y' = 4x^3 + 12x^2$
- b $y'' = 12x^2 + 24x$
- c $(0, 2), (-3, -25)$
- d $(0, 2)$ is a point of inflection, and $(-3, -25)$ is a minimum point.
- e $(0, 2), (-2, -14)$
- f $(0, 2)$ is a horizontal point of inflection, while $(-2, -14)$ is an ordinary point of inflection.
- g $x < -2$ or $x > 0$

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- a $f'(x) = (x - 5)(3x - 21)$
- b $(5, 0), (7, -4)$
- c $f''(x) = 6x - 36$
- d Local minimum at $(7, -4)$, local maximum at $(5, 0)$
- e $x < 6$

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- a
 - i $(3, -61), (-3, 47)$
 - ii $(-3, 47)$ is a maximum turning point and $(3, -61)$ is a minimum turning point.
- b
 - i $(2, -18), (-2, 14)$
 - ii $(-2, 14)$ is a maximum turning point and $(2, -18)$ is a minimum turning point.
- c
 - i $(-4, 112), (1, -13)$
 - ii $(-4, 112)$ is a maximum turning point and $(1, -13)$ is a minimum turning point
- d
 - i $(-1, 2), (0, 3), (1, 2)$
 - ii $(0, 3)$ is a maximum turning point, $(-1, 2)$ and $(1, 2)$ are minimum turning points

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a

i $x = -\frac{3}{7}, -\frac{1}{3}$

ii Maximum turning point at $x = -\frac{3}{7}$ and minimum turning point at $x = -\frac{1}{3}$

b

i $x = 1, 7$

ii Maximum turning point at $x = 1$ and minimum turning point at $x = 7$

c

i $x = -2$

ii Maximum turning point.

d

i $x = 5$

ii Horizontal point of inflection

e

i $x = 5$

ii Horizontal point of inflection.

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a

$$x = -\frac{3}{5}, -\frac{1}{3}$$

b

$$x = -\frac{7}{15}$$

c

$$2$$

25

a

$$y'' = 30(x^2 - 5)(x^2 - 1)$$

b

$$(-\sqrt{5}, 0), (-1, -64), (1, -64), (\sqrt{5}, 0)$$

c

$$(-\sqrt{5}, 0), (\sqrt{5}, 0)$$

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a

i $(5, 4)$

ii $(5, 4)$

iii Turning point. The graph changes from decreasing to increasing on either side of $x = 5$, but the concavity does not change at $x = 5$.

b

i $(4, 5)$

ii $(4, 5)$

iii The graph changes from decreasing to increasing at $x = 4$, but the concavity does not change at $x = 4$.

c

i $(8, 6)$

ii $(8, 6)$

iii The graph changes from decreasing to increasing around $x = 8$, but the concavity does not change around $x = 8$.

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a $a = b$



b $a = b$

c They can be any real numbers.

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a $3a + 2b + 12 = 0$

b $b = -3a$

c $a = 4$

d $b = -12$

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a $a = -2$

b $c = -48$